Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

Ben Tang | Greg Cogan Lab, Duke University

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Technical design document for the Neural Field Perceiver. Claims are labeled **supported by related work** (published evidence in adjacent settings, not directly validated for intra-op μECoG), **internally observed** (our experiments), or **working hypothesis** (unverified). Structural assumptions that could be wrong are marked in red.

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1 Problem

Each patient's μECoG array is a sparse spatial sample of cortical activity during speech. The array sits on left ventral sensorimotor cortex (vSMC), and its position in MNI-152 space is determined post-hoc from electrode reconstruction (availability for our cohort is conditional — see Section 8).

Input. For patient i with N_i electrodes at MNI positions $\{\mathbf{p}_j^{(i)}\}$, we observe high-gamma analytic amplitude (HGA, 70–150 Hz) at 200 Hz:

$$d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t) \tag{1}$$

where $a(\mathbf{p}, t)$ is the cortical HGA field and ϵ captures noise, impedance variation, and sub-electrode activity.

Output. Infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$ from a 9-phoneme vocabulary — three separate 9-way classifications per non-word token (e.g., /a/, /b/, /e/).

Cross-patient challenge. Electrode j on patient i and electrode j on patient i' sample different cortical locations. Arrays are offset 15–25 mm in MNI with variable rotation. Channel indices are meaningless across patients.

Constraint	Value	Impact
Electrodes/patient	63–201	Sparse cortical sampling
Electrode spacing	∼2 mm	Near-columnar resolution
Cross-patient offset	15–25 mm	Different regions sampled
Trials/patient	46–178	Small-data regime
Utterance/patient	∼1 min	Too little for SSL
Phonemes	9 classes × 3 positions	Low-dimensional output

2 Key Idea: Shared Neural Field + Reprojection

2.1 Decomposition

Cross-patient decoding requires shared structure. We decompose each patient's cortical field:

$$a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t) \tag{2}$$

- f: shared neural field — the spatiotemporal pattern phoneme $\mathbf{y}$ produces in canonical cortical space.
- T_i: patient-specific spatial transform (estimated by MNI normalization, imperfect).
- $r^{(i)}$: patient-specific residual (individual neural organization, signal quality, pathology).

Cross-patient decoding works to the extent that $\|f\| \gg \|r^{(i)}\|$.

2.2 The factorization

The shared field f factorizes through a low-dimensional bottleneck:

$$d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + r^{(i)}\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t) \tag{3}$$

where:

- $g(\mathbf{p}, \mathbf{z})$: the **somatotopic map** — given a brain position and an articulatory state, what is the neural activity? Time-invariant: lip motor cortex encodes lip state the same way at t=0 and t=500 ms. Absorbs all spatial dependence.
- $\mathbf{z}(t) \in \mathbb{R}^k$ (k≈5–8): the **articulatory state** — tongue position, lip aperture, voicing, etc. Absorbs all time and phoneme dependence.
- $\dot{\mathbf{z}}(t) = h(\mathbf{z}(t); \mathbf{y})$: dynamics on the articulatory manifold.

Space and time interact only through the k-dimensional bottleneck $\mathbf{z}(t)$. Instead of learning every (position × time × phoneme) combination, the model learns a spatial basis g and temporal dynamics h composed through $\mathbf{z}$. This is what makes the problem tractable with 63–201 electrodes.

The somatotopic map is static. g does not change during a trial. Cross-patient alignment is therefore a static geometric problem: different patients have electrodes at different positions sampling the same map. The temporal dynamics $\mathbf{z}(t)$ are shared across patients — the same phoneme produces similar trajectories regardless of who speaks it.

2.3 The 3D vision analogy

This structure maps to multi-view 3D reconstruction:

3D Reconstruction	**Neural Field Decoding**
3D body model	Shared somatotopic map $g(\mathbf{p}, \mathbf{z})$
Pose parameters	Articulatory state $\mathbf{z}(t)$
Camera extrinsics	MNI offset $\Delta^{(i)}$, rotation $\omega^{(i)}$
Camera intrinsics	Per-patient gain/bias
2D keypoints	Electrode observations $d_j^{(i)}(t)$
Reprojection loss $\|\pi(V) - x\|^2$	**Reprojection loss** $\|g(\mathbf{p}, \hat{\mathbf{z}}) - d\|^2$

Reprojection loss: if your latent state estimate is correct, you should predict what every electrode observes. Self-supervised but geometrically grounded — unlike JEPA/VICReg/masked reconstruction, the target is physically meaningful.

Where the analogy breaks. (1) Each patient is a different "scene," not a different camera on the same scene — the shared structure is weaker than photometric consistency. (2) We have 63–201 electrodes, not 65K pixels — the reprojection signal per frame is thin. (3) The "scene" changes every frame. These mean reprojection is a useful regularizer, not the primary signal. The supervised classification loss remains primary.

2.4 Evidence and central hypothesis

Central hypothesis. *Intra-operative μECoG speech recordings share enough spatiotemporal structure across patients that a coordinate-aware spatial encoder, trained with reprojection supervision, will improve cross-patient phoneme decoding beyond methods that treat each patient's electrode array as unrelated.* This is testable via ablation A1 (Section 6). The document designs an architecture to test this hypothesis; it does not assume the hypothesis is true.

Supported by related work. Per-patient input layers + shared backbone transfers in adjacent settings: Singh [1] (N=25 sEEG, PER 0.49 group vs. 0.57 individual); Willett [2] (per-day affine + shared GRU). Articulatory somatotopy in vSMC is reproducible [3, 4]. Motor manifold structure is

preserved across individuals [5, 6]. MNI coordinates improve cross-patient ECoG transfer: Chen [7] (SwinTW, PCC 0.765 on unseen patients).[1]

Internally observed. LOPO achieves PER = 0.762 on S14 (frozen backbone, grouped-by-token CV) — above chance (0.89) but worse than per-patient (0.737). 55 experiments converge to PER 0.750–0.780, suggesting a systematic barrier (data scarcity, alignment, or both — not yet isolated). Articulatory bottleneck head improves LOPO decoding, consistent with f organized along articulatory dimensions.

3 Architecture

The encoder maps electrode observations and positions to a shared spatial representation:

$$s^{(i)}(t) = \text{Encoder}\big(\{d_j^{(i)}(t),\ \tilde{\mathbf{p}}_j^{(i)}\}_{j=1}^{N_i}\big) \in \mathbb{R}^{L\times d} \tag{4}$$

with L=16 virtual electrodes and $\tilde{\mathbf{p}}_j^{(i)}$ the corrected MNI position. The observation decoder then predicts each electrode's activity from its local latent:

$$\boxed{\hat{d}_j^{(i)}(t) = g_\theta\big(\tilde{\mathbf{p}}_j^{(i)},\ s_j^{*,(i)}(t),\ e^{(i)}\big)} \tag{5}$$

where $s_j^{*,(i)}(t) = \sum_\ell a_{j\ell} \cdot s_\ell^{(i)}(t)$ is a position-weighted aggregation of virtual electrode outputs at electrode j's location (Section 3.7), and $e^{(i)}$ is a patient embedding.

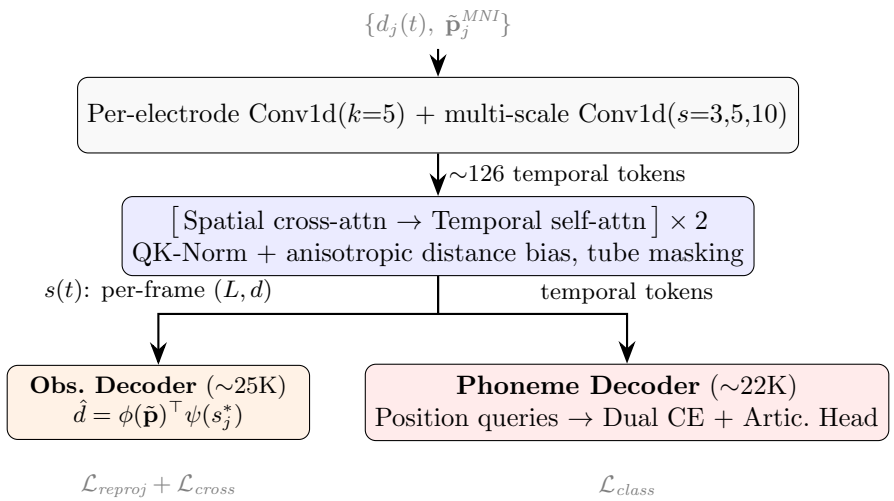
Structural identifiability. The encoder sees **no patient identity** — only electrode positions and activity. The patient embedding $e^{(i)}$ enters only the observation decoder, where it handles patient-specific signal characteristics (impedance, gain, tissue properties). This is a deliberate architectural constraint: by withholding patient identity from the encoder, the latent $s^{(i)}(t)$ is forced to represent the neural field in a patient-agnostic frame, encoding spatial structure through MNI coordinates rather than arbitrary per-patient codes.

Residual leakage: Even without $e^{(i)}$, the encoder implicitly identifies patients through their unique electrode position sets. This is inherent to any coordinate-aware architecture and is the *intended* behavior — position-dependent processing is the whole point. The constraint prevents only *non-geometric* patient-specific processing. Whether this is sufficient for practical patient-invariance is tested by ablation A3 (comparing with e in the encoder) and A5 (reprojection weight).

The losses create a gradient tension on the *decoder*: $\mathcal{L}_{\text{reproj}}$ (with $e^{(i)}$) rewards patient-specific reconstruction; $\mathcal{L}_{\text{cross}}$ (with e=$\mathbf{0}$) rewards patient-invariant spatial structure; $\mathcal{L}_{\text{class}}$ rewards whatever predicts phonemes.

[1] Standard ECoG (10 mm spacing), not μECoG (2 mm). None of these results use intra-operative recordings, μECoG resolution, or our small-data regime. They support the general direction, not the specific formulation.

3.1 Overview



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda_1 \mathcal{L}_{\text{reproj}} + \lambda_2 \mathcal{L}_{\text{cross}}, \qquad \lambda_1\text{: } 0.5\rightarrow 0.1, \quad \lambda_2\text{: } 0.3\rightarrow 0.05} \tag{6}$$

Factorized interleaved processing replaces strict spatial→temporal separation. Each layer alternates spatial cross-attention (virtual electrodes aggregate from real electrodes within a time window) and temporal self-attention (each virtual electrode attends across time windows). The spatial distance bias is **static** — the same at every time step — respecting the time-invariance of g (Section 2.2). Temporal attention captures dynamics in $\mathbf{z}(t)$. Interleaving lets temporal context denoise spatial estimates at every layer.

Two representations:

- $s(t) \in \mathbb{R}^{L\times d}$: virtual electrode outputs after spatial cross-attention. Used by the observation decoder for reprojection.
- $\mathbf{z}_p \in \mathbb{R}^d$: per-phoneme-position readout via learned query cross-attention over temporal tokens. Used for classification.

3.2 Electrode Tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \rightarrow d)}_{\text{activity}} + \underbrace{\text{Linear}\big(\gamma(\tilde{\mathbf{p}}_j) \rightarrow d\big)}_{\text{Fourier MNI PE}} \tag{7}$$

where $\tilde{\mathbf{p}}_j = R(\omega^{(i)}) \cdot \mathbf{p}_j + \Delta^{(i)}$ is the corrected MNI position. **No patient embedding enters the encoder** — patient identity is withheld to enforce structurally patient-agnostic representations (Section 3).

Fourier positional encoding. Following NeRF [17], **standardized** MNI coordinates (zero-mean, scaled so the speech motor region spans $[-1, 1]$ per axis) pass through multi-scale sinusoidal features before projection:

$$\gamma(\bar{\mathbf{p}}) = \big[\sin(2^0\pi\bar{\mathbf{p}}),\ \cos(2^0\pi\bar{\mathbf{p}}),\ \ldots,\ \sin(2^{F-1}\pi\bar{\mathbf{p}}),\ \cos(2^{F-1}\pi\bar{\mathbf{p}})\big] \in \mathbb{R}^{6F} \tag{8}$$

with F=4 frequency levels. MNI coordinates are standardized **before** encoding: the speech motor region spans ∼30 mm per axis, so $\bar{\mathbf{p}} \in [-1, 1]$ maps 1 unit ≈ 15 mm. The resulting wavelengths are $\lambda_\ell = 2/2^\ell$ in standardized units: ℓ=0 → 30 mm (full-array), ℓ=1 → 15 mm (articulatory region), ℓ=2 → 7.5 mm (sub-region), ℓ=3 → 3.75 mm (near electrode spacing). A linear projection Linear(24 → d) maps to token dimension.[2]

Per-patient encoder parameters (6 per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: translation (init $\mathbf{0}$). Corrects placement offset.
- $\omega^{(i)} \in \mathbb{R}^3$: axis-angle rotation (init $\mathbf{0}$, Rodrigues formula). Corrects orientation.

These are purely geometric — they adjust electrode positions, not processing. The patient embedding $e^{(i)} \in \mathbb{R}^d$ (init $\sim\mathcal{N}(0, 0.02)$) enters only the observation decoder (Section 3.7), where $\phi(\tilde{\mathbf{p}}, e)$ can learn position-dependent patient corrections.

Residual risk: $\Delta^{(i)}$ and $\omega^{(i)}$ could in principle serve as arbitrary per-patient codes rather than geometric corrections. Regularization (L2 on Δ, ω; small init) limits this channel, and 6 scalar parameters are a narrow information bottleneck compared to the d-dimensional e. Ablation A3 tests the contribution of each per-patient parameter.

3.3 Temporal Preprocessing

Per-electrode denoising. Conv1d(k=5, stride=1) on each electrode's time series provides ∼25 ms temporal smoothing.

Multi-scale tokenization. Three parallel Conv1d branches at strides 3, 5, 10 (sampling intervals 15, 25, 50 ms) on each denoised electrode time series. The branches provide three temporal resolutions; what temporal structure each one captures depends on the kernel size, downstream attention, and data — not on the stride alone. Tokenization uses the production window $[0, T_{\text{utt}}]$ (≈1.0 s, 200 samples at 200 Hz): stride 3 yields 66 tokens, stride 5 yields 40, stride 10 yields 20, totaling ≈126 temporal tokens per electrode. Concatenated with learnable per-scale embeddings + sinusoidal temporal PE (encoding each token's center time in seconds).

3.4 Virtual Electrodes

$L = 16$ learned query vectors at learned MNI positions. Fixed model parameters, the same for every patient and time step. During spatial cross-attention, each virtual electrode attends to all real electrodes, weighted by spatial proximity.

After cross-attention, virtual electrode ℓ holds a d-dimensional summary of cortical activity near its position. The output $s(t) \in \mathbb{R}^{L \times d}$ is a spatially-indexed representation in a shared frame — every patient maps to the same 16 positions regardless of electrode count or grid layout.

Atlas initialization. Virtual electrode positions are initialized at Brainnetome atlas [19] centroids of speech motor subregions: lip M1, tongue M1, jaw M1, laryngeal M1, lip S1, tongue S1, ventral premotor, supplementary motor area, and adjacent somatosensory zones. This gives anatomically meaningful starting positions so backprop only needs to fine-tune, not discover cortical organization

[2] NeRF uses F=10 for sub-pixel detail. We use F=4 because (a) ∼200 electrode positions across ∼20 patients cannot constrain higher frequencies, and (b) the finest wavelength (3.75 mm) already approaches electrode spacing (∼2 mm) and inter-subject correspondence uncertainty (∼5 mm). The ℓ=3 frequency is useful *within-patient* (where MNI is precise for rigid arrays) but unreliable for *cross-patient* alignment at that scale. Ablation A12 tests F=3 (dropping the finest frequency).

from scratch. Repulsive regularization $\mathcal{L}_{\text{repel}} = \sum_{i \neq j} \max(0, r_{\min} - \|\text{v_pos}_i - \text{v_pos}_j\|)$ prevents collapse.

Why 16. 63–201 electrodes cover $\sim 20 \times 30$ mm. 16 virtual electrodes gives ~ 5 mm spacing, roughly matching somatotopic scale. Ablation A4 tests $\{4, 8, 16, 32\}$.

Coverage gaps. Virtual electrodes in uncovered cortical regions receive only weak, distant attention and will drift toward covered regions or produce noise. The union of all patients' coverage is larger than any single patient's — more patients and cross-task pooling expand the coverage envelope. No architecture fixes missing data.

3.5 Interleaved Spatiotemporal Blocks ($\times 2$)

Each block alternates two attention operations:

(a) Spatial cross-attention (per temporal window). Virtual electrodes (Q) attend to real electrode tokens (K, V) within each time window. Attention logits use **QK-Norm + anisotropic distance bias**:

$$A_{ij} = \tau_h \cdot \frac{Q_i \cdot K_j}{\|Q_i\| \, \|K_j\|} \; - \; (\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j)^\top \Sigma_i^{-1} (\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j) \tag{9}$$

where $\tau_h > 0$ is a learnable temperature per head (init $\sqrt{d_h}$, clamped ≤ 100), $\bar{\mathbf{p}}$ denotes standardized MNI coordinates, and $\Sigma_i = C_i C_i^\top$ is a per-virtual-electrode covariance (C_i: lower-triangular Cholesky factor, 6 params each, init αI). Normalizing Q and K to unit vectors keeps the content term bounded, so the distance bias has a consistent effect throughout training regardless of embedding norms — following SwinV2 [24] and ViT-22B. Each virtual electrode learns the **shape** of its spatial receptive field: an elongated lip region, a compact tongue area, etc. 96 + 4 params (16×6 covariances + 4 temperatures). 4-head attention.

The distance bias is **static**: Σ_i depends only on virtual electrode identity, not on time or input content. The spatial geometry of attention is the same at every time step, consistent with g being time-invariant. What changes with time are the **values** (electrode activity), not the spatial weighting structure.

(b) Temporal self-attention (per virtual electrode). Each of the 16 virtual electrodes attends across all $\sim$126 temporal tokens. 4 heads, sinusoidal temporal PE + per-scale embeddings. This captures articulatory dynamics $h(\mathbf{z}; \mathbf{y})$.

Optional 1-layer spatial self-attention between virtual electrodes within each time window (captures inter-zone co-articulation, e.g., simultaneous lip rounding and tongue backing).

Block internals. Each sub-layer uses Pre-LayerNorm: $\text{out} = x + \text{SubLayer}(\text{LN}(x))$. The FFN after each attention operation is SwiGLU [25]:

$$\text{FFN}(x) = \big(xW_1 \odot \text{SiLU}(xW_{\text{gate}})\big) W_2, \qquad W_1, W_{\text{gate}} \in \mathbb{R}^{d \times d_{\text{ff}}}, \; W_2 \in \mathbb{R}^{d_{\text{ff}} \times d} \tag{10}$$

with $d_{\text{ff}} = \lfloor 8d/3 \rfloor$, matching the parameter count of a standard $4d$ ReLU FFN. Dropout p=0.2 on attention weights and FFN output. DropPath p=0.1 on the residual stream. Each block thus contains: LN $\rightarrow$ spatial cross-attn $\rightarrow$ LN $\rightarrow$ FFN $\rightarrow$ LN $\rightarrow$ temporal self-attn $\rightarrow$ LN $\rightarrow$ FFN.

Tube masking. Instead of masking electrodes independently per frame, mask each selected electrode across a contiguous time window of 5–10 frames (25–50 ms). Masking is **scale-consistent**: when electrode j is masked at time t, it is masked at all three temporal resolutions simultaneously.

Otherwise the model trivially reconstructs masked information from a finer or coarser scale — the somatotopic map $g(\mathbf{p}, \mathbf{z})$ is the same physical function at every temporal resolution. This forces interpolation in **both** space (from nearby electrodes at the same time) **and** time (from the same electrode at adjacent windows), exercising both types of shared structure.

Masking curriculum. Start at 30%, anneal to 70% over training. Low masking early helps the spatial encoder learn basic distance-activity relationships. High masking later forces robust spatial interpolation. The somatotopic map is spatially smooth ($\sim$5 mm scale), so 70% masking is physically feasible.

Spatial readout. Attention pooling (not MaxPool) preserves the somatotopic decomposition:

$$s_{\text{pool}}(t) = \sum_{i=1}^{L} w_i \cdot \text{out}_i(t), \qquad w_i = \text{softmax}\big(\mathbf{q}_{\text{read}}^{\top} \text{out}_i(t)/\sqrt{d}\big) \tag{11}$$

Per-position readout via learned queries. Three learned query vectors $\mathbf{q}_p \in \mathbb{R}^d$ ($p \in \{1, 2, 3\}$) cross-attend to the full spatially-pooled temporal token sequence:

$$\mathbf{z}_p = \text{CrossAttn}\big(\mathbf{q}_p,\ K{=}\{s_{\text{pool}}(t)\}_t,\ V{=}\{s_{\text{pool}}(t)\}_t\big) \tag{12}$$

Each query discovers where its phoneme's information concentrates in the temporal sequence, accounting for variable phoneme durations, coarticulation across boundaries, and CVC/VCV structural asymmetry (VCV: 0.68 s vs. CVC: 0.58 s [10]). Inspired by DETR's object queries [23], which learn to attend to task-relevant regions without hard spatial partitioning. Ablation A13 compares against fixed equal-window averaging. Global readout $z_{\text{global}} = \text{mean}(\text{all temporal tokens})$ for k-NN.

3.6 Why Perceiver, not ViT or point-cloud methods?

ViT requires a fixed regular grid. Our patients have different grid sizes (8×16, 12×22) at different MNI positions. Cannot use.

Point-cloud methods (PointNet [15], Point Transformer [16]) handle irregular 3D point sets and are legitimate alternatives. Perceiver cross-attention is simpler for our case: shared query vectors directly provide the fixed-size shared output we need, and our 63–201 points are too small for hierarchical architectures. Worth testing as an ablation.

3.7 Observation Decoder ($\sim$25K params)

The forward model — predicts what each electrode should observe given the latent state. Structurally, this is a DeepONet [18]: a branch network encodes the latent field, a trunk network encodes spatial position, and they interact through a low-rank inner product. The neural operator literature validates this as the right inductive bias for sparse field reconstruction.

Position-weighted aggregation. Each electrode j gets a local latent by weighting virtual electrode outputs by spatial proximity:

$$s_j^*(t) = \sum_{\ell=1}^{L} a_{j\ell} \cdot \text{out}_\ell(t), \qquad a_{j\ell} = \frac{\exp\big(-\alpha_{\text{dec}} \cdot \|\bar{\tilde{\mathbf{p}}}_j - \text{v_}\bar{\text{p}}\text{os}_\ell\|^2\big)}{\sum_m \exp\big(-\alpha_{\text{dec}} \cdot \|\bar{\tilde{\mathbf{p}}}_j - \text{v_}\bar{\text{p}}\text{os}_m\|^2\big)} \tag{13}$$

Bilinear prediction.

$$\hat{d}_j^{(i)}(t) = \phi(\tilde{\mathbf{p}}_j,\ e^{(i)})^\top\ \psi\big(s_j^*(t)\big) \tag{14}$$

$\phi = \mathrm{MLP}_{\mathrm{spatial}}(\tilde{\mathbf{p}}, e) \in \mathbb{R}^k$: spatial tuning (trunk). $\psi = \mathrm{MLP}_{\mathrm{latent}}(s_j^*) \in \mathbb{R}^k$: latent activation (branch). k=32.

The decoder is the **only** component that sees patient identity. $\phi(\tilde{\mathbf{p}}, e)$ learns position-dependent patient effects — partially modeling the nonlinear residual $\delta_i(\mathbf{p})$. This asymmetry (patient-agnostic encoder, patient-aware decoder) is the architectural mechanism that enforces identifiability (Section 3). Ablation A9 tests a full MLP fallback.

Within-patient reprojection loss.

$$\mathcal{L}_{\mathrm{reproj}} = \frac{1}{N_i T} \sum_{t,j} \big\| d_j(t) - \hat{d}_j(t) \big\|^2 \tag{15}$$

Computed over all electrodes including masked ones (the model must interpolate).

Cross-patient reprojection loss. During multi-patient training, when patients A and B have trials with the same token (e.g., both producing /abe/), the shared observation function should explain patient B's observations from patient A's latent state. Both sides are averaged over the **full trial**:

$$\mathcal{L}_{\mathrm{cross}} = \frac{1}{N_B} \sum_{j=1}^{N_B} \big\| \bar{d}_j^{(B)} - g_\theta(\tilde{\mathbf{p}}_j^{(B)},\ \bar{s}_A^{*,(B)},\ \mathbf{0}) \big\|^2 \tag{16}$$

where $\bar{d}_j^{(B)} = \frac{1}{T}\sum_t d_j^{(B)}(t)$ is patient B's trial-averaged observation, and $\bar{s}_A^{*,(B)} = \frac{1}{T}\sum_t s_A^{*,(B)}(t)$ aggregates patient A's trial-averaged latent at patient B's electrode positions. **The patient embedding is set to 0** — the decoder must predict patient B's observations using only spatial position and patient A's latent, with no patient-identity shortcut.

$\mathcal{L}_{cross}$ is a deliberately coarse spatial-alignment signal. Trial-level averaging collapses exactly the temporal structure that the encoder's learned queries and multi-scale tokenization are designed to preserve. This is an intentional tradeoff, not an oversight:

- $\mathcal{L}_{\mathrm{class}}$ captures **fine temporal** structure (per-position phoneme classification via learned queries).
- $\mathcal{L}_{\mathrm{reproj}}$ captures **fine spatiotemporal** structure (per-frame, per-electrode, within-patient).
- $\mathcal{L}_{\mathrm{cross}}$ captures **coarse spatial** alignment (trial-averaged, cross-patient).

Full-trial averaging is also a practical necessity: per-phoneme onset times from MFA alignment are unreliable for ∼50% of patients [9], which motivated full-trial CTC over per-position classification in our existing pipeline. Using phoneme-window averaging in $\mathcal{L}_{\mathrm{cross}}$ would reintroduce the same unreliable timing.

Same-token pairing limitation. Trials are matched by identical token (e.g., /abe/ ↔ /abe/). With 52 tokens and multiple patients, this provides $\sim O(K^2 \cdot 52)$ pairs from K patients. However, same-token matching means the model can satisfy $\mathcal{L}_{\mathrm{cross}}$ by learning token-level templates without discovering a general spatial field. The encoder's lack of patient identity and the decoder's e=**0** constraint limit this escape hatch, but do not eliminate it. This is the weakest of the three losses and is expected to serve as a regularizer, not the primary learning signal. $\lambda_2 < \lambda_1$ and anneals aggressively; the λ_2=0 fallback drops it entirely (ablation A5).

Optional: Beta-NLL. Replace MSE with learned per-electrode variance to downweight noisy electrodes:

$$\mathcal{L}_{\text{reproj}}^{\text{UQ}} = \frac{1}{2}\Big(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2}\Big) \cdot \big(\hat{\sigma}_j^2\big)_{\text{sg}}^{0.5} \tag{17}$$

3.8 Phoneme Decoder (∼5K classification + ∼17K readout)

Dual head per position $p \in \{1, 2, 3\}$:

$$\text{logits}_p = \frac{1}{2}\big(\underbrace{\text{Linear}(z_p \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \text{normalize}(\text{Linear}(z_p \to 15)) \cdot A^\top}_{\text{articulatory head}}\big) \tag{18}$$

$A \in \mathbb{R}^{9\times 15}$: fixed articulatory feature matrix. τ: learnable temperature. Separate heads per position. $\mathcal{L}_{\text{class}} = \frac{1}{3}\sum_p \text{FocalCE}(\text{logits}_p, y_p;\ \gamma{=}2, \text{LS}{=}0.1)$ with mixup (α=0.2).

3.9 Parameter Budget

Module	**Params**	**Shared?**	**Role**
Interleaved blocks (×2)	∼160K	Shared	Spatial inversion + dynamics
incl. anisotropic $\Sigma_i + \tau_h$	100	Shared	Receptive field shape + QK temp
incl. virtual positions	48	Shared	Atlas-initialized MNI coords
Position queries + cross-attn	∼17K	Shared	Learned phoneme readout
Observation decoder	∼25K	Shared	Reprojection forward model
Phoneme dual heads (×3)	∼5K	Shared	Classification
Multi-scale Conv1d	∼12K	Shared	Temporal tokenization
$\Delta^{(i)} + \omega^{(i)}$ (encoder)	6/pt	Per-patient	Geometric correction
$e^{(i)}$ (decoder only)	64/pt	Per-patient	Patient-specific reconstruction
Total shared	**∼219K**		
Grand total (10 pts)	**∼220K**		(cf. current: 122K)

4 Training and Evaluation

4.1 Phase 1: Source training

$N-1$ source patients (where N is the total PS cohort; exact count depends on LOPO target). Each batch:

1. Construct tokens with Fourier MNI PE. Apply tube masking (curriculum 30→70%).
2. Interleaved blocks: spatial cross-attn → temporal self-attn ×2.
3. Within-patient reprojection on all electrodes (including masked) → $\mathcal{L}_{\text{reproj}}$.
4. Cross-patient reprojection on same-token trial pairs → $\mathcal{L}_{\text{cross}}$.
5. Per-position readout → dual head → $\mathcal{L}_{\text{class}}$.
6. $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda_1\mathcal{L}_{\text{reproj}} + \lambda_2\mathcal{L}_{\text{cross}}$.

AdamW with differential LR: shared params 10^{-3}; geometric params (Δ, ω, virtual positions, Σ) 3×10^{-4}; patient e at 10^{-3}. Linear warmup (20 epochs) $\rightarrow$ cosine decay. Weight decay 10^{-4}, grad clip 5.0, batch 16. Early stopping on held-out 20% source validation, patience 7.

4.2 Phase 2: Target adaptation (LP-FT)

Per fold of 5-fold grouped-by-token CV on target patient:

1. **Linear probe.** Freeze encoder, train decoder + $\Delta^{(\text{target})} + \omega^{(\text{target})} + e^{(\text{target})}$. 150 epochs. Best checkpoint by inner validation (20% held out from training fold).
2. **Fine-tune.** Unfreeze encoder at 0.1× LR. 100 epochs, patience 7.

4.3 Phase 3 (optional): Transductive reprojection pre-training

Pre-train spatial encoder + decoder using $\mathcal{L}_{\text{reproj}}$ only. If target patient's unlabeled data is included, this is transductive and must be reported separately from strict LOPO.

4.4 Evaluation regimes

1. **Zero-shot.** Phase 1 only. $\Delta = \mathbf{0}$, $\omega = \mathbf{0}$, $e = \mathbf{0}$, all frozen. Pure coordinates-only generalization — no patient-specific parameters at all. Strongest claim, likely worst PER.
2. **LP-FT (primary).** Phase 1 $\rightarrow$ Phase 2. Matches clinical calibration scenario.
3. **Transductive.** Phase 3 $\rightarrow$ Phase 1 $\rightarrow$ Phase 2. Reported separately.

Metrics. PER (edit distance), grouped-by-token 5-fold CV. Full recipe: weighted k-NN (k=10, cosine), TTA 16, dual head, source k-NN at 0.5× weight. Conformal prediction sets as interpretability output.

5 What Our Experiments Show

5.1 Evaluation recipe matters more than architecture

Component added	Category	PER	Δ
Baseline (CE, mean-pool)	–	0.872	–
+ Label smoothing 0.1	Training	0.831	−4.1 pp
+ Per-position heads + dropout	Training	0.819	−1.2 pp
+ Mixup α=0.2	Training	0.802	−1.7 pp
+ Weighted k-NN (k=10)	Evaluation	0.781	−2.1 pp
+ TTA (16 copies)	Evaluation	0.767	−1.4 pp
+ Articulatory + CE dual head	Model	0.737	−3.0 pp

Cumulative marginal gains, not independent. Total: −13.5 pp. All on S14, **per-patient** training, grouped-by-token CV. Per-fold noise is ±0.05 PER, so improvements < 2 pp may not replicate. The dual head's −3.0 pp is the marginal effect of adding articulatory + CE on top of the previous recipe.

5.2 Multi-scale temporal: largest isolated LOPO architectural gain

In **LOPO** experiments, stride 3+5+10 gave PER 0.766 $\rightarrow$ 0.750 (−1.6 pp), the largest *isolated single-change* gain from any architectural modification (distinct from the cumulative per-patient

gains in the table above). However, this is below the ± 5 pp per-fold noise floor and needs multi-seed replication to confirm. Validated on BiGRU; needs re-validation on transformer.

5.3 The $\sim$0.76 convergence

55 LOPO experiments converge to PER 0.750–0.780. Not attributable to any single bottleneck: S14 is no longer truly held out (55+ architectures evaluated); $\sim$30 samples/fold give ± 0.05 noise; S14 has only 153 trials; spatial alignment is one factor among several.

5.4 SSL failed at this data scale

All SSL methods near chance (PER 0.85–0.91). JEPA + LP-FT: 0.797. With $\sim$1 min utterance/patient, SSL cannot learn useful representations from HGA.

6 Ablation Plan

ID	Ablation	Question
A1	Coordinate-aware spatial encoding	**Central**: do coordinates help?
	a. Current Conv2d+BiGRU baseline	$\rightarrow$ 0.762 (anchor)
	b. Perceiver, **no coordinate PE** (uniform attn)	$\rightarrow$ architecture benefit alone
	c. Perceiver + Fourier MNI PE	$\rightarrow$ coordinate benefit (c vs b)
	d. (c) + reprojection loss	$\rightarrow$ reprojection benefit (d vs c)
	e. (d) + cross-patient reproj.	$\rightarrow$ cross-patient benefit (e vs d)
	f. (c) + random positions (sabotage)	$\rightarrow$ sanity check: MNI > random
A2	Population-level (all patients)	Is the 0.76 wall S14-specific?
A3	Per-patient params	Δ only / $+\omega$ / $+e_{\text{decoder}}$ / $+e_{\text{encoder}}$ (identifiability test)
A4	Virtual electrode count	$L \in \{4, 8, 16, 32\}$
A5	Reprojection weight	$\lambda_1, \lambda_2 \in \{0, 0.01, 0.1, 0.5\}$
A6	Cross-task pooling	+ Duraivel 2025 data
A7	Reprojection loss variant	Beta-NLL vs. MSE
A8	Spatial attention	Isotropic α vs. anisotropic Σ_i
A9	Bilinear vs. MLP decoder	$\phi^\top \psi$ vs. MLP($[\tilde{\mathbf{p}}; e; s_j^*]$)
A10	Interleaved vs. separated	Interleaved blocks vs. strict spatial$\rightarrow$temporal
A11	Masking strategy	Tube vs. per-frame; masking ratio 30–80%
A12	Positional encoding variants	$\gamma(\mathbf{p})$ vs. Linear($\mathbf{p} \rightarrow d$) vs. F=3 (drop 3.75 mm)
A13	Position readout	Learned queries vs. fixed equal-window averaging

7 Priorities and Risks

7.1 Priority ordering

The document's own evidence suggests:

1. **Cross-task data pooling (A6).** Lowest risk. Same phonemes, overlapping patients [10]. Directly addresses data scarcity. Works with current pipeline and new architecture.
2. **Population-level evaluation (A2).** Pure methodology fix. Required to distinguish S14-specific effects from universal trends. Must be the primary success criterion.

3. **MNI coordinate architecture (A1, this document).** Highest novelty, highest risk. The hypothesis that spatial alignment is the binding constraint is plausible but unvalidated — it has not been isolated from data scarcity or measurement noise.

7.2 Success criteria

Primary: Mean PER across all patients improves over current LOPO baseline by ≥ 3 pp (≥ 3 seeds, LP-FT regime, paired Wilcoxon per-patient test at α=0.05). The 3 pp threshold is pre-registered because our per-fold noise is ± 5 pp and per-patient noise across seeds is ± 2 pp; improvements below this cannot be reliably distinguished from measurement error. MNI benefit isolated by paired comparisons: A1c vs. A1b (coordinate value, same architecture) and A1d vs. A1c (reprojection value).

Interpretability (weak evidence): Virtual electrode positions cluster over known speech motor regions — but since positions are atlas-initialized, this partly reflects initialization bias, not learned structure. Stronger evidence: positions that *move away* from atlas priors in a somatotopically coherent direction. Spatial tuning vectors $\phi(\mathbf{p})$ recover articulatory somatotopy. Learned $\Delta^{(i)}$ and $\omega^{(i)}$ correlate with known registration quality.

7.3 Risks

Risk	Severity	Mitigation
MNI coordinates unavailable	High	Fallback to scaled grid positions (A1b tests no-coordinate baseline; A1f tests random positions as sabotage control)
MNI alignment too imprecise	High	Ablation A1; $\Delta^{(i)}/\omega^{(i)}$ correct residual misalignment; F=3 variant (A12) avoids finest frequency
Reprojection competes with classification	Med	λ ablation A5; λ=0 fallback
Perceiver overfits (200K, 1300 trials)	Med	Masking; early stopping; reproj. reg.
Bilinear separability too restrictive	Med	Ablation A9: MLP fallback
Rigid (Δ, ω) too crude for surgical brains	Med	$e^{(i)}$ absorbs non-geometric residuals; if insufficient, test pos-dep e (A3)
Virtual electrodes collapse	Low	Atlas init; repulsive regularization

8 Available Data

Cohort	Patients	Trials	MNI?	Usage
PS (Spalding published)	8 patients†	$\sim$1096	Expected*	Source / target
PS (additional)	S16, S57†	$\sim$250	Expected*	Source
Cross-task (Duraivel 2025)	3 overlapping pts	156–208/pt	Expected*	Priority 1
Lexical task	13 patients	Varies	Expected*	Future source
Total unique PS	**10 patients**	**$\sim$1350**		

†S32 = S36 (duplicate, counted once). 12 PS entries in BIDS $\rightarrow$ 11 unique $\rightarrow$ 10 usable (S18 excluded). The 8 Spalding patients include S14 (153 trials), used as Phase 2 target in pilot results. For LOPO, source = $N-1$ patients.

*MNI coordinates available from BioImage Suite (DBS) / Brainlab (tumor) outputs on Box.[3]

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